



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Linear Equations & Inequalities

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Your notes

Solving Linear Equations

Solving Linear Equations

What are linear equations?

- A **linear equation** is one that can be written in the form $ax + b = c$
 - a , b , and c are **numbers** and x is the **variable**
 - $2x + 3 = 5$
 - $3x + 4 = 1$
 - $x - 5 = -3$
- The greatest power of x is 1
 - There are no terms like x^2

How do I solve linear equations?

- You need to use **operations** like adding, subtracting, multiplying and dividing to get x on its **own**
- Any operation you do to **one side** of the **equation** must also be done to the **other side**
- For example, to solve $2x + 1 = 9$ look at the $+1$ on the left
 - Undo this by subtracting 1 from **both sides** and **simplifying**

$$2x + 1 = 9$$

 (-1) (-1)

$$2x = 8$$

- This equation is now **easier** to solve
- $2x$ is $2 \times x$ so undo this by dividing **both** sides by 2 and **simplifying**

$$2x = 8$$

 $(\div 2)$ $(\div 2)$

$$x = 4$$

- The **solution** to the equation is $x = 4$



Your notes

- Adding 1 was undone by subtracting 1
- Multiplying by 2 was undone by dividing by 2
 - **Addition** and **subtraction** are said to be **inverse** (opposite) operations
 - **Multiplication** and **division** are also **inverse** operations

Does the order of steps matter?

- As long as each step is applied correctly, the order in which inverse operations are applied **does not matter**
 - Applying the operations in one order may be easier than another
- Consider $4x + 8 = 12$
 - It is **easier** to **first subtract** 8 from both sides

$$4x = 4$$

- **Then divide** both sides by 4

$$x = 1$$

- If you want to first divide by 4, a common **mistake** is to write $x + 8 = 3$
 - This is **incorrect** as 8 has not been divided by 4
 - You must divide **every term** by 4

$$\frac{4x}{4} + \frac{8}{4} = \frac{12}{4}$$
$$x + 2 = 3$$

- Then subtract 2 from both sides

$$x = 1$$

How do I solve linear equations with negative numbers?

- For example, $2 - 3x = 10$
 - Subtract 2 from both sides and simplify

$$2 - 3x = 10$$

$$(-2)$$

$$(-2)$$



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$$-3x = 8$$

- Then divide both sides by -3 and simplify

$$-3x = 8$$

$$(\div -3)$$

$$(\div -3)$$

$$x = -\frac{8}{3}$$

- Some people prefer to write $2 - 3x = 10$ as $-3x + 2 = 10$, which is equivalent
 - You then subtract 2 and divide by -3 as before



Examiner Tips and Tricks

Substitute your answer back into the original equation to check you got it right!



Worked Example

Solve the equation

$$9 - 7x = 5$$

Subtract 9 from both sides of the equation

$$-7x = -4$$

Divide both sides by -7

Remember that a negative divided by a negative will result in a positive number

$$x = \frac{-4}{-7}$$

$$x = \frac{4}{7}$$

How do I solve linear equations with brackets?



Your notes

- If a linear equation involves brackets, **expand** the brackets first

- For example, solve $2(x - 3) = 10$

- Expand the brackets

$$2x - 6 = 10$$

- Then solve as shown previously

- Add 6 then divide by 2

$$2x = 16$$

$$x = 8$$

- **Expanding** brackets **first** will always work, but you can also divide first

- Dividing **both** sides of $2(x - 3) = 10$ by 2 gives $(x - 3) = 5$

- which gives $x = 8$

- This method works but can lead to harder **fractions**

How do I solve linear equations with fractions?

- If a linear equation contains **fractions**, **multiply both sides** by the **lowest common denominator**

- For example, $\frac{x}{5} + 4 = \frac{9}{2}$

- The lowest common denominator of 5 and 2 is 10

- Multiply **all terms** on both sides by 10

$$\left(10 \times \frac{x}{5}\right) + (10 \times 4) = 10 \times \frac{9}{2}$$

$$\frac{10x}{5} + 40 = \frac{90}{2}$$

- **Simplify** the fractions

$$2x + 40 = 45$$

- Now solve as before, by subtracting 40, then dividing by 2



Your notes

$$2x = 5$$

$$x = \frac{5}{2}$$

- Unless the question specifies otherwise, you can leave the answer like this
 - A **decimal** or **mixed number** would also be **accepted**

What if the unknown is on the denominator?

- For example $\frac{4}{x-2} = 3$

- Multiply both sides of the equation by the denominator

$$\frac{4}{x-2} \times (x-2) = 3(x-2)$$

- Simplify the fractions, and expand any brackets

$$4 = 3(x-2)$$

$$4 = 3x - 6$$

- Now solve as before, by adding 6 to both sides, then dividing by 3

$$10 = 3x$$

$$\frac{10}{3} = x$$



Worked Example

(a) Solve the equation

$$5(3 - 4x) + 1 = 26$$

Expand the bracket

$$15 - 20x + 1 = 26$$

Simplify

$$16 - 20x = 26$$



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It can be helpful to consider $16 - 20x$ as $-20x + 16$

Subtract 16 from both sides

$$-20x = 10$$

Divide both sides by -20 and simplify

$$x = \frac{10}{-20}$$

$$x = -\frac{1}{2}$$

$$x = -\frac{1}{2}$$

-0.5 is also accepted

(b) Solve the equation

$$\frac{5x}{4} = \frac{1}{2}$$

The lowest common denominator of 4 and 2 is 4

Multiply both sides by 4

$$4 \times \frac{5x}{4} = 4 \times \frac{1}{2}$$

Simplify (cancel) the fractions

$$5x = 2$$

To solve this equation, divide both sides by 5

$$x = \frac{2}{5}$$

0.4 is also accepted

How do I solve linear equations with x terms on both sides?

- **Collect the x terms** (or whichever variable is involved) **together on one side**
 - To do this, **remove all the x terms from one side**



Your notes

- It is easiest to remove the **smallest x term** to avoid negatives
- For example, $4x - 7 = 11 + x$
- Remove the x term on the right-hand side, by subtracting x from both sides

$$4x - 7 = 11 + x$$

$$\begin{array}{ccc} & (-x) & (-x) \\ 4x - 7 & = & 11 + x \end{array}$$

$$3x - 7 = 11$$

- There are no longer any x terms on the right
- This now has the same form as previously seen
 - Solve by adding 7 then dividing by 3

$$3x - 7 = 11$$

$$3x = 18$$

$$x = 6$$



Worked Example

Solve the equation

$$4 - 5x = 6x - 29$$

Remove the x terms from either side

We will remove them from the left as $-5x$ is smaller than $6x$

Add $5x$ to both sides

$$4 = 11x - 29$$

Get $11x$ on its own by adding 29 to both sides

$$33 = 11x$$

Divide both sides by 11 to find x

$$3 = x$$

$$\mathbf{x = 3}$$



Your notes

Solving Linear Inequalities

Solving Linear Inequalities

What is an inequality?

- An **inequality** tells you that something is **greater than** ($>$) or **less than** ($<$) something else
 - $x > 5$ means x is greater than 5
 - x could be 6, 7, 8, 9, ...
- Inequalities may also include being **equal** ($=$)
 - $\geq$ means **greater than or equal to**
 - $\leq$ means **less than or equal to**
 - $x \leq 10$ means x is less than or equal to 10
 - x could be 10, 9, 8, 7, 6, ...
- When they **cannot** be **equal**, they are called **strict** inequalities
 - $>$ and $<$ are **strict** inequalities
 - $x > 5$ **does not** include 5 (strict)
 - $x \geq 5$ **does** include 5 (not strict)

How do I find integers that satisfy inequalities?

- You may be given **two end points** and have to list the **integer** values of x that **satisfy** the inequality
 - Look at whether each end point is **included** or not
 - $3 \leq x \leq 6$
 - $x = 3, 4, 5, 6$
 - $3 \leq x < 6$
 - $x = 3, 4, 5$
 - $3 < x \leq 6$
 - $x = 4, 5, 6$
 - $3 < x < 6$



Your notes

- $x = 4, 5$
- If only **one end point** is given, there are an infinite number of integers
 - $x > 2$
 - $x = 3, 4, 5, 6, \dots$
 - $x \leq 2$
 - $x = 2, 1, 0, -1, -2, \dots$
 - Remember **zero** and **negative** whole numbers are **integers**
 - If the question had said **positive** integers only then just list $x = 2, 1$
- You may be asked to find integers that satisfy **two** inequalities
 - $0 < x < 5$ and $x \geq 3$
 - List **separately**: $x = 1, 2, 3, 4$ and $x = 3, 4, 5, 6, \dots$
 - Find the values that appear in **both** lists: $x = 3, 4$
- If the question does not say x is an integer, do **not assume** x is an integer!
 - $x > 3$ actually means **any value** greater than 3
 - 3.1 is possible
 - $\pi = 3.14159\dots$ is possible
- You may be asked to find the **smallest** or **largest** integer
 - The smallest integer that satisfies $x > 6.5$ is 7



Worked Example

List all the integer values of x that satisfy

$$-4 \leq x < 2$$

Integer values are whole numbers

$-4 \leq x$ shows that x includes -4 , so this is the first integer

$$x = -4$$

$x < 2$ shows that x does not include 2

Therefore the last integer is $x = 1$



Your notes

$$x = 1$$

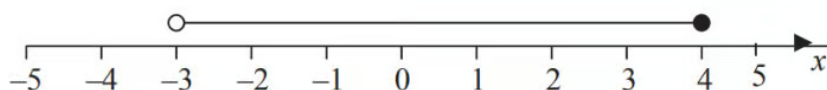
For the answer, list all the integers from -4 to 1

Remember integers can be zero and negative

$$x = -4, -3, -2, -1, 0, 1$$

How do I represent an inequality on a number line?

- The inequality $-3 < x \leq 4$ is shown on a **number line** below



- Draw **circles** above the **end points** and connect them with a **horizontal line**

- Leave an **open circle** for end points with **strict** inequalities, $<$ or $>$

- These end points **are not** included

- Fill in a **solid circle** for end points with $\leq$ or $\geq$ inequalities

- These end points **are** included

$<$ or $>$ use an open circle ○ (end points are excluded)

$\leq$ or $\geq$ use a closed circle ● (end points are included)

- Use a **horizontal arrow** for inequalities with **one end point**

- $x > 5$ is an open circle at 5 with a horizontal arrow pointing to the right



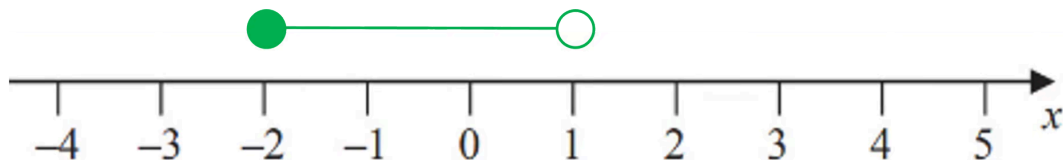
Worked Example

Represent the following inequalities on a number line.

(a) $-2 \leq x < 1$

-2 is included so use a closed circle

1 is not included so use an open circle

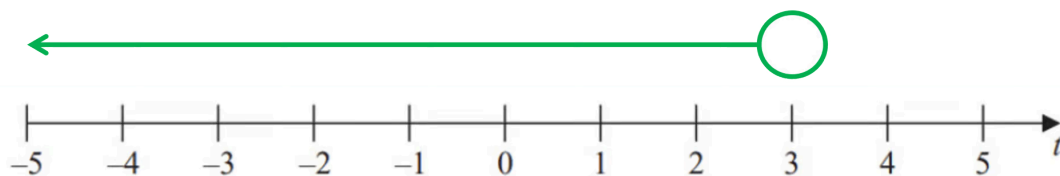


(b) $t < 3$

3 is not included so use an open circle

There is no second end point

Any value less than three is accepted, so draw a horizontal arrow to the left



How do I solve a linear inequalities?

- Solving linear inequalities is just like **Solving Linear Equations**
 - Follow the same rules, but keep the **inequality sign** throughout
 - If you change the inequality sign to an equals sign you are changing the meaning of the problem
- When you **multiply** or **divide** both sides by a **negative** number, you must **flip the sign** of the inequality

E.g.

$$\begin{array}{ccc} & 1 < 2 \\ (\times -1) & & (\times -1) \\ -1 > -2 \end{array}$$

- Never multiply** or **divide** by a **variable** (x) as this could be **positive or negative**
- The **safest way** to rearrange is simply to **add** and **subtract** to move all the terms onto **one side**

How do I solve double inequalities?



Your notes

- Inequalities such as $a < 2x < b$ can be solved by doing the same thing to **all three parts** of the inequality
 - Use the same rules as solving linear inequalities



Examiner Tips and Tricks

- Do **not** change the inequality sign to an equals when solving linear inequalities.
 - In an exam you will lose marks for doing this.
- Remember to **reverse the direction** of the inequality sign when multiplying or dividing by a **negative** number!



Worked Example

Solve the inequality $2x - 5 \leq 21$.

Add 5 from both sides

$$2x \leq 26$$

Now divide both sides by 2

$$x \leq 13$$

$$\mathbf{x \leq 13}$$



Worked Example

Solve the inequality $5 - 2x \leq 21$.

Subtract 5 from both sides, keeping the inequality sign the same

$$-2x \leq 16$$

Now divide both sides by -2.

However because you are dividing by a negative number, you must flip the inequality sign

$$x \geq -8$$

$$x \geq -8 \text{ or } -8 \leq x$$



Your notes